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Anther: Upper swollen part with microsporangia.

Filament: Stalk of stamen

Connective: Tissue connecting anther lobes with filament

Anther typically contains two compartments called **thecae** (singular theca). Each theca consists of two microsporangia. Two microsporangia fused to form a **locule.**

Sterile stamens are called **Staminodes**. Example: *Cassia*. **Distinct:** stamens which do not fuse to one another. **Free:** stamens which do not fuse with other parts of flower. **Apostemonous:** flowers with stamens that are free and distinct.

4.4.1 Fusion of stamens: Refers to the stamens fusing among themselves or with other parts of flower. Two types.

1. Connation and 2. Adnation

1. Connation: Refers to the fusion of stamens among themselves. It is of 3 types. **a.** Adelphy. **b.** Syngenecious. **c.** Synandrous.

a. Adelphy: Filaments connate into one or more bundles but anthers are free. It may be the following types.

1. Monadelphous: Filaments of stamens connate into a single bundle. Example: malvaceae (chinarose,cotton).



Figure 4.24: (a)
Monadelphous

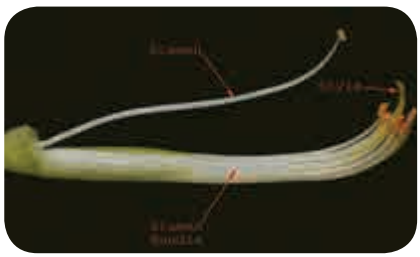


Figure 4.24: (b)
Diadelphous



Figure 4.24: (c)
Polyadelphous

2. Diadelphous: Filaments of stamens connate into two bundles. Example: Fabaceae, pea.

3. Polyadelphous: Filaments connate into many bundles. Example: *Citrus*, *Bombax*

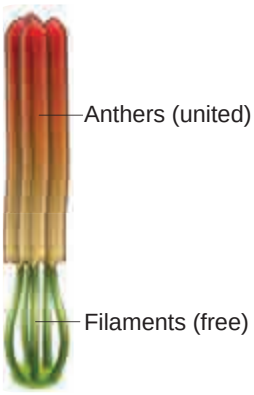


Figure 4.24: (d)
Syngenesious



Figure 4.24: (e)
Synandrous

b. Syngenesious: Anthers connate, filaments free. Example: Asteraceae.

c. Synandrous: Filaments and anthers are completely fused. Example: *Coccinea*.

2. Adnation: Refers to the fusion of stamens with other floral parts. **Epipetalous** (petalostemonous): Stamens are adnate to petals .Example: brinjal,*Datura*.

a. Episepalous: stamens are adnate to sepals. Example: *Grevillea* (Silver oak)

b. Epitepalous (epiphyllous): stamens are adnate to tepals. Example: *Asphodelus*, *Asparagus*.

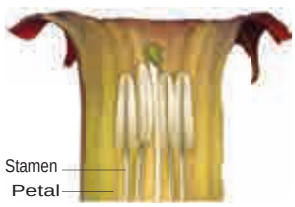


Figure 4.25: (a)
Epipetalous